Fiscal Year:	FY 2010	**Task Last Updated:**	FY 05/25/2011
PI Name:	Hlatky, Lynn Ph.D.		
Project Title:	NSCOR: Solid Tumor Risk Estimation: Incorporating Intercellular Interaction Effects		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02135-2907	**Congressional District:**	8
Comments:	PI affiliation name change to Steward Research and Specialty Projects Corporation (2012). PI formerly at Caritas St. Elizabeth's Medical Center, Boston, MA (2005-2011); Dana-Farber Cancer Institute, Boston, MA (prior to 2005). Subsequent name change from Steward Research and Specialty Projects Corporation to GeneSys Research Institute in 2013.		
Project Type:	Ground	**Solicitation / Funding Source:**	2004 NSCOR Space Radiation NNH04ZUU002N
Start Date:	10/01/2004	**End Date:**	10/31/2010
No. of Post Docs:	10	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	12
No. of Bachelor's Candidates:	12	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: End date should be 10/31/2010, per M. Klumpar in PI's office [Ed., 3/28/2011] NOTE: Received NCE to 3/31/2011 per C. Guidry/JSC (10/2010) NOTE: Received NCE to 9/30/2010 per J. Dardano/JSC (9/09) NOTE: New grant number NNJ06HA28G received in 2006 when PI moved to Tufts and end date is 9/30/2009 (information received 6/2009) jvp NOTE: End date changed back to 9/30/2009 per S. Krenek/JSC (8/07) End date changed to 10/31/2009 per JSC info update (10/06)		
Key Personnel Changes/Previous PI:	Dr J. Folkmann deceased		

COI Name (Institution):	Folkman, Judah M.D. (Children's Hospital, Harvard Medical School) Hahnfeldt, Philip Ph.D. (Caritas St. Elizabeth's Medical Center, Tufts Univ. School of Medicine) Huber, Peter (German Cancer Research Center (dkfz)) Lamont, Clare (Caritas St. Elizabeth's Medical Center, Tufts Univ. School of Medicine) Sachs, Rainer Ph.D. (University of California)
Grant/Contract No.:	NNJ06HA28G ; NNJ04HJ12G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	We will improve solid tumor risk estimation for astronauts. Improvements will be based in large part on analyzing data resulting from our high-energy iron ion and proton in vivo irradiations. Our mathematical risk modeling will use standard computational approaches which minimize theoretical assumptions and also use minimally-parameterized, biologically-based carcinogenesis models capable of interrelating epidemiological data, animal experiments, and in vitro radiobiology. In our experiments and our computational analyses, special emphasis will be placed on tumor promotion and progression, as influenced by intercellular signaling among various cell populations within the tumor (e.g. tumor cells, endothelial cells and stroma). The in vivo models used are highly suitable for the program. These include the transgenic conditional k-ras lung tumor model from the Tyler Jacks's laboratory that will allow the study of radiation modulation of all steps in carcinogenesis. These mice have a recombinant adenovirus expressing Cre recombinase to induce k-ras expression which synchronizes tumor onset, allowing numerous tumors to develop simultaneously in the lungs. The number of tumors and rate as which they progress though the four stages of carcinogenesis (e.g. time-dependent ratio of hyperplastic to neoplastic lesions) can then be traced as a function of the timing, dose and quality of the radiation. We will also use dormant human breast tumor xenograft models which allow us to place particular emphasis on dissecting the progression stage of carcinogenesis. These in vivo models offer specific advantages: solid tumor sites of concern to NASA (lung, breast, and later thyroid) will be studied; middle-age mice will be used; and due to the nature of these models comparatively few mice will be needed and post-irradiation study times will be comparatively short. In this Program we implement a multi-level approach with whole organism-, tissue- cell-, and molecular-level endpoints used to measure radiation response. For examples, in addition to radiation influence on latency periods or the time-dependent ratio of dysplastic versus frankly neoplastic lung lesions, we examine cell signaling, use matrix and clustering computer algorithms for analyzing transcriptome data, and interpret chromosome aberrations scored with mFISH/SKY using our established computer simulation software. Importantly for our emphasis on intercellular signaling as a key aspect of carcinogenesis, we will assay not only tumor cells but also tumor-associated stromal and endothelial cells (including circulating endothelial cells). It is turning out that such tissues can help support or repress tumors (i.e. play a major role in promotion and progression), and due to the relative stability of their genomes they are simpler to analyze for radiation-induced DNA damage. An experimental and theoretical emphasis on tumor progression is also planned, based on the fact that this step in carcinogenesis has hitherto received less attention from radiation risk modelers than other steps, e.g. initiation, but is at least as important. There is now strong evidence that microscopic dormant neoplastic sites are far more prevalent in adults than previously assumed and that their progression can be accelerated by radiation. Radiation shortening of latency periods could thus be a key component of solid tumor risk for middle-aged astronauts. A tightly-knit interdisciplinary team, Director L. Hlatky, Associate Director R. Sachs along with the other Project leaders J. Folkman, P. Huber, and P. Hahnfeldt will carry out the 5 closely interrelated Projects: (1) mouse models for assessing carcinogenesis risk; (2) HZE and low LET irradiation; (3) radiation transcriptome analyses; (4) quantitative chromosome aberration analysis; and (5) quantitative estimation of solid tumor risk. Project (5) will integrate data from the first four Projects as well as drawing on results from the literature. It will culminate in a composite radiation carcinogenesis model designed to reduce the uncertainties in risk estimates for astronauts by addressing cell interaction effects and including in the multi-step carcinogenesis analysis the overlooked step of progression.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Autopsy findings from adults dying from non-cancer causes have now unequivocally established that most of us possess fully-malignant tumors that are held in check by a balance between cell cycling and cell death. Such tumors are characterized by a failure to advance through the angiogenic switch. Dormant tumors that manage to switch to the angiogenic phenotype grow exponentially and ultimately become clinically apparent, which is referred to as the "disease state" of cancer (Folkman, Nature 2004). As recent results from Project 2 disclose, major angiogenic genes show significant regulation by space radiations. This suggests there may be a link between space radiation exposure and modulation of the dormancy bottleneck that holds existing indolent lesions in check. To research the matter, the development of dormant tumor models has been a priority. We now have in place several human dormancy models in mice. The models have been phenotypically characterized (Almog and Folkman, JNCI 2006 and FASEB 2004). Tumors remain dormant for a prolonged period of time until they spontaneously switch to the fast-growing phenotype. The time point of the switch is tumor-type dependent. We recovered several of the spontaneously-switched tumors from mice and cultured them in vitro. Re-implantation in naive mice confirmed a stable fast-growing phenotype. It is concluded a genetic reprogramming event accompanies transition from dormancy to fast-growth. We have designed new methods for estimating the incidence of iatric (i.e. treatment induced) cancers after radiotherapy. As cancer radiotherapy methods improve and the number of patients treated increases, more and more people survive long enough after the initial cancer diagnosis to potentially develop radiotherapy-induced second malignancies. Because treatment techniques are changing rapidly, there is insufficient follow-up time to directly measure the second cancer risks associated with the newer methods. Biologically-based mathematical models, which implement plausible assumptions about radiation carcinogenesis processes such as initiation of organ-specific stem cells to the pre-malignant state, inactivation (killing), and repopulation of both normal and pre-malignant stem cells during radiotherapy and

	subsequent organ recovery, can address this problem: Models calibrated with older data can predict second cancer risks of new/prospective radiotherapy protocols.
Task Progress:	Our funded NSCOR focused on mechanistic distillation of progression events critical to radiation-induced carcinogenesis (as per the original RFA). The focus was lung cancer, but many findings apply to other radiation-induced cancers of interest to NASA. We made many findings at BNL pertinent to the high priority topics set forth by the NASA Space Radiation Program. We also ushered in several pivotal concepts, now accepted as basic precepts in the field of space radiation carcinogenesis. Selected Key Highlights * Recognized that radiation-induced cancer types (e.g. lung, breast, thyroid) are prevalent in even healthy individuals, including astronauts, but in a microsopic dormant state. Formulated and introduced the concept that radiation-carcinogenesis risk is not just the risk of initiation, and transformation events making new cancer cells, but also about cancer cells already present progressing to manifest disease. * Found proton and 56 Fe oppositely regulate major signaling pathways critical in radiation-carcinogenesis progression, including angiogenesis. * Demonstrated proton suppression of the angiogenesis bottleneck in carcinogenesis which acts to reduce this component of risk. * Found age-at-exposure effects for HZE modulation of mouse lung cancer progression. * Found and confirmed unexpected differences in molecular signaling responses to protons and gamma-rays that we continue to explore at the biophysical level. * Found the endothelial cell genome is more strongly regulated in response to proton radiation while fibroblasts are more significantly regulated by 56 Fe. * Discovered dose-dependent behavior of 56 Fe irradiation promoted passage though both promotion and progression carcinogenesis phases. * Demonstrated cell-cell exchange between primary non-transformed cells can increase the radiation resistance of the cells to both low LET (proton, gamma) and high LET (Fe, C) radiation. * Demonstrated space radiation exposure elicits both pro- and anti-carcinogenic longterm effects. * Developed a biologically-based two-stage logistic (TSL) model for early radiation carcinogenesis that anticipates the influence of growth-limiting effects at the population level into the promotion of radiation-initiated cells. It has certain predictive advantages over the current standard Two-Stage Clonal Expansion model. * Mathematically determined that cell-cell interactions modify the basic cell kinetics of tumor progression in a manner that predicts a nonlinear dose dependencies for radiogenic cancer risk. [Ed. note (5/25/2011): FY2010 technical report submitted in January 2011; FY2010 Task Book report provided in May 2011]
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